

# Course Machine Learning

## Course #6 – Naïve Bayes Classification

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## 1 — Naïve Bayes Classification

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The Naive Bayesian classifier is based on Bayes' theorem with the independence assumptions between predictors. A Naive Bayesian model is easy to build, with no complicated iterative parameter estimation which makes it particularly useful for very large datasets. Despite its simplicity, the Naive Bayesian classifier often does surprisingly well and is widely used because it often outperforms more sophisticated classification method.

### *Algorithm*

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Bayes theorem provides a way of calculating the posterior probability,  $P(c|x)$ , from  $P(c)$ ,  $P(x)$ , and  $P(x|c)$ . Naive Bayes classifier assume that the effect of the value of a predictor ( $x$ ) on a given class ( $c$ ) is independent of the values of other predictors. This assumption is called class conditional independence.

$$P(c|x) = \frac{P(x|c)P(c)}{P(x)}$$

Likelihood
Class Prior Probability  
Posterior Probability
Predictor Prior Probability

$$P(c|X) = P(x_1|c) \times P(x_2|c) \times \dots \times P(x_n|c) \times P(c)$$

- $P(c|x)$  is the posterior probability of *class (target)* given *predictor (attribute)*.
- $P(c)$  is the prior probability of *class*.
- $P(x|c)$  is the likelihood which is the probability of *predictor* given *class*.
- $P(x)$  is the prior probability of *predictor*.

In ZeroR model there is no predictor, in OneR model we try to find the single best predictor, naive Bayesian includes all predictors using Bayes' rule and the independence assumptions between predictors.

### What is a Naive Bayes Classifier?

The Naive Bayes algorithm is a supervised machine learning algorithm based on the Bayes' theorem. It is a probabilistic classifier that is often used in NLP tasks like sentiment analysis (identifying a text corpus' emotional or sentimental tone or opinion).

The Bayes' theorem is used to determine the probability of a hypothesis when prior knowledge is available. It depends on conditional probabilities. The formula is given below :

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

where  $P(A|B)$  is posterior probability i.e. the probability of a hypothesis A given the event B occurs.  $P(B|A)$  is likelihood probability i.e. the probability of the evidence given that hypothesis A is true.  $P(A)$  is prior probability i.e. the probability of the hypothesis before observing the evidence and  $P(B)$  is marginal probability i.e. the probability of the evidence. When the Bayes' theorem is applied to classify text documents, the class c of a particular document d is given by :

$$c_{MAP} = \operatorname{argmax}_{c \in C} P(c | d)$$

$$= \operatorname{argmax}_{c \in C} \frac{P(d | c)P(c)}{P(d)}$$

$$= \operatorname{argmax}_{c \in C} P(d | c)P(c)$$

$$= \operatorname{argmax}_{c \in C} P(x_1, x_2, \dots, x_n | c)P(c)$$

MAP is "maximum a posteriori" = most likely class

Bayes Rule

Dropping the denominator

Document d represented as features  $x_1..x_n$

Let the feature conditional probabilities  $P(X_i|C)$  be independent of each other (conditional independence assumption).

### Prior and Posterior Probability

Bayes' Theorem describes the probability of an event, based on a prior knowledge of conditions that might be related to that event.

$$P(H|E) = \frac{P(E|H) * P(H)}{P(E)}$$

Likelihood of the Evidence given that the Hypothesis is True (points to  $P(E|H)$ )  
Prior Probability of the Hypothesis (points to  $P(H)$ )  
Posterior Probability of the Hypothesis given that the Evidence is True (points to  $P(H|E)$ )  
Prior Probability that the evidence is True (points to  $P(E)$ )

### What makes Naive Bayes a "Naive" algorithm?

Naive Bayes classifier assumes that the features we use to predict the target are independent and do not affect each other. While in real-life data, features depend on each other in determining the target, but this is ignored by the Naive Bayes classifier.

Though the independence assumption is never correct in real-world data, but often works well in practice. so that it is called "**Naive**".

### Math behind Naive Bays Algorithm

Given a features vector  $X=(x_1, x_2, \dots, x_n)$  and a class variable  $y$ , Bayes Theorem states that:

$$P(y|X) = \frac{P(X|y) * P(y)}{P(X)}$$

We're interested in calculating the posterior probability  $P(y | X)$  from the likelihood  $P(X | y)$  and prior probabilities  $P(y), P(X)$ .

Using the chain rule, the likelihood  $P(X | y)$  can be decomposed as:

$$\begin{aligned} P(X|y) &= P(x_1, x_2, \dots, x_n|y) \\ &= P(x_1|x_2, \dots, x_n, y) * P(x_2|x_3, \dots, x_n, y) \dots P(x_n|y) \end{aligned}$$

but because of the Naive's conditional independence assumption, the conditional probabilities are independent of each other.

$$P(X|y) = P(x_1|y) * P(x_2|y) \dots P(x_n|y)$$

Thus, by conditional independence, we have:

$$P(y|X) = \frac{P(x_1|y) * P(x_2|y) \dots P(x_n|y) * P(y)}{P(x_1) * P(x_2) \dots P(x_n)}$$

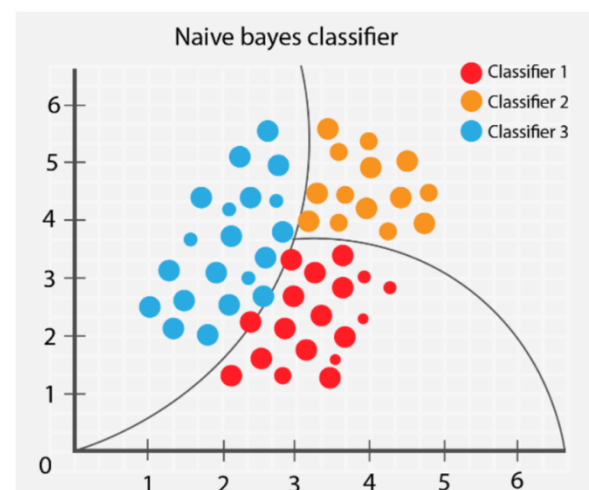
And as denominator remains constant for all values, the posterior probability can then be:

$$P(y|x_1, x_2, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i|y)$$

The Naive Bayes classifier combines this model with a decision rule. One common rule is to pick the hypothesis that's most probable; this is known as the maximum a posteriori or MAP decision rule.

$$y = \operatorname{argmax}_y P(y) \prod_{i=1}^n P(x_i|y)$$

We can illustrate these results by the flowing figure.



## Types of Naive Bayes Classifiers

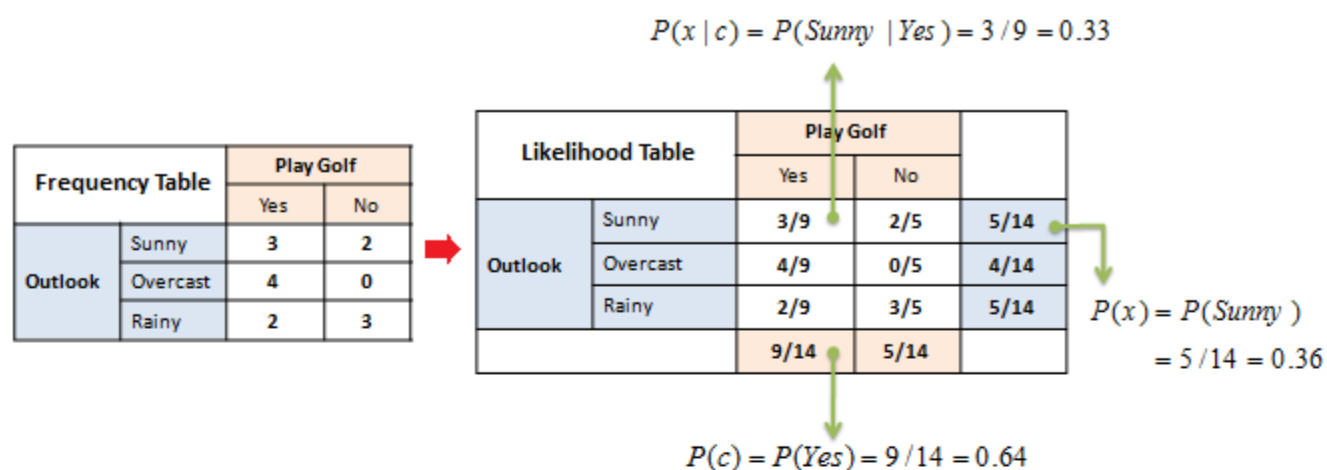
- **Multinomial:** Feature vectors represent the frequencies with which certain events have been generated by a multinomial distribution. For example, the count how often each word occurs in the document. This is the event model typically used for document classification.
- **Bernoulli:** Like the multinomial model, this model is popular for document classification tasks, where binary term occurrence(i.e. a word occurs in a document or not) features are used rather than term frequencies(i.e. frequency of a word in the document).
- **Gaussian:** It is used in classification and it assumes that features follow a normal distribution.

## Playing Golf Naïve Bayes Classification Example

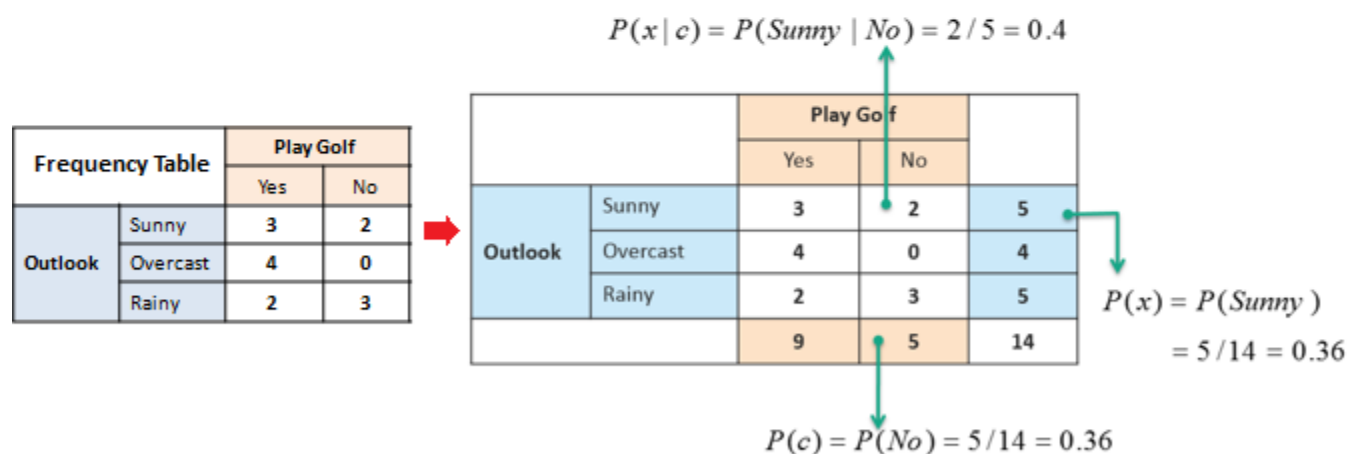
We use the same simple Weather dataset here.

| Outlook  | Temp | Humidity | Windy | Play Golf |
|----------|------|----------|-------|-----------|
| Rainy    | Hot  | High     | False | No        |
| Rainy    | Hot  | High     | True  | No        |
| Overcast | Hot  | High     | False | Yes       |
| Sunny    | Mild | High     | False | Yes       |
| Sunny    | Cool | Normal   | False | Yes       |
| Sunny    | Cool | Normal   | True  | No        |
| Overcast | Cool | Normal   | True  | Yes       |
| Rainy    | Mild | High     | False | No        |
| Rainy    | Cool | Normal   | False | Yes       |
| Sunny    | Mild | Normal   | False | Yes       |
| Rainy    | Mild | Normal   | True  | Yes       |
| Overcast | Mild | High     | True  | Yes       |
| Overcast | Hot  | Normal   | False | Yes       |
| Sunny    | Mild | High     | True  | No        |

The posterior probability can be calculated by first, constructing a frequency table for each attribute against the target. Then, transforming the frequency tables to likelihood tables and finally use the Naive Bayesian equation to calculate the posterior probability for each class. The class with the highest posterior probability is the outcome of prediction.



Posterior Probability:  $P(c | x) = P(\text{Yes} | \text{Sunny}) = 0.33 \times 0.64 \div 0.36 = 0.60$



Posterior Probability:  $P(c | x) = P(\text{No} | \text{Sunny}) = 0.40 \times 0.36 \div 0.36 = 0.40$

The likelihood tables for all four predictors.

Frequency Table

|         |          | Play Golf |    |
|---------|----------|-----------|----|
|         |          | Yes       | No |
| Outlook | Sunny    | 3         | 2  |
|         | Overcast | 4         | 0  |
|         | Rainy    | 2         | 3  |



Likelihood Table

|         |          | Play Golf |     |
|---------|----------|-----------|-----|
|         |          | Yes       | No  |
| Outlook | Sunny    | 3/9       | 2/5 |
|         | Overcast | 4/9       | 0/5 |
|         | Rainy    | 2/9       | 3/5 |

|          |        | Play Golf |    |
|----------|--------|-----------|----|
|          |        | Yes       | No |
| Humidity | High   | 3         | 4  |
|          | Normal | 6         | 1  |



|          |        | Play Golf |     |
|----------|--------|-----------|-----|
|          |        | Yes       | No  |
| Humidity | High   | 3/9       | 4/5 |
|          | Normal | 6/9       | 1/5 |

|       |      | Play Golf |    |
|-------|------|-----------|----|
|       |      | Yes       | No |
| Temp. | Hot  | 2         | 2  |
|       | Mild | 4         | 2  |
|       | Cool | 3         | 1  |



|       |      | Play Golf |     |
|-------|------|-----------|-----|
|       |      | Yes       | No  |
| Temp. | Hot  | 2/9       | 2/5 |
|       | Mild | 4/9       | 2/5 |
|       | Cool | 3/9       | 1/5 |

|       |       | Play Golf |    |
|-------|-------|-----------|----|
|       |       | Yes       | No |
| Windy | False | 6         | 2  |
|       | True  | 3         | 3  |



|       |       | Play Golf |     |
|-------|-------|-----------|-----|
|       |       | Yes       | No  |
| Windy | False | 6/9       | 2/5 |
|       | True  | 3/9       | 3/5 |

### Prediction of an event by naïve bayes

In this example we have 4 inputs (predictors). The final posterior probabilities can be standardized between 0 and 1.

| Outlook | Temp | Humidity | Windy | Play |
|---------|------|----------|-------|------|
| Rainy   | Cool | High     | True  | ?    |

$$P(Yes | X) = P(Rainy | Yes) \times P(Cool | Yes) \times P(High | Yes) \times P(True | Yes) \times P(Yes)$$

$$P(Yes | X) = 2/9 \times 3/9 \times 3/9 \times 3/9 \times 9/14 = 0.00529 \rightarrow 0.2 = \frac{0.00529}{0.02057 + 0.00529}$$

$$P(No | X) = P(Rainy | No) \times P(Cool | No) \times P(High | No) \times P(True | No) \times P(No)$$

$$P(No | X) = 3/5 \times 1/5 \times 4/5 \times 3/5 \times 5/14 = 0.02057 \rightarrow 0.8 = \frac{0.02057}{0.02057 + 0.00529}$$

### The zero-frequency problem

Add 1 to the count for every attribute value-class combination (*Laplace estimator*) when an attribute value (*Outlook=Overcast*) doesn't occur with every class value (*Play Golf=no*).

### Numerical Predictors

Numerical variables need to be transformed to their categorical counterparts (**binning**) before constructing their frequency tables. The other option we have is using the distribution of the numerical variable to have a good guess of the frequency. For example, one common practice is to assume normal distributions for numerical variables.

The probability density function for the normal distribution is defined by two parameters (mean and standard deviation).

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{Mean}$$

$$\sigma = \left[ \frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2 \right]^{0.5} \quad \text{Standard deviation}$$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{Normal distribution}$$

## 2 — Naïve Bayes Classifier Step by Step

### *Understanding Naive Bayes Classifier/Prediction using Bayes Theorem*

Let's see step by step how Naive Bayes will work for the following problem statement. **Predict whether a person will buy a product given the specific day, discount and free delivery using Naive Bayes Classifier.**

Here is the data, which comprises of Day which can be Weekday/Weekend/Holiday, Discount can be Yes/No, Free delivery can be Yes/No.

| Day     | Discount | Free Delivery | Buy |
|---------|----------|---------------|-----|
| Weekday | Yes      | Yes           | Yes |
| Weekday | Yes      | Yes           | No  |
| Weekday | No       | No            | No  |
| Holiday | Yes      | Yes           | Yes |
| Weekend | Yes      | Yes           | Yes |
| Holiday | Yes      | Yes           | Yes |
| Weekend | Yes      | No            | Yes |
| Weekday | Yes      | Yes           | No  |
| Weekend | Yes      | Yes           | Yes |
| Holiday | Yes      | Yes           | Yes |
| Holiday | No       | Yes           | Yes |
| Holiday | No       | No            | No  |
| Weekend | Yes      | Yes           | Yes |
| Holiday | Yes      | Yes           | Yes |
| Weekend | Yes      | Yes           | Yes |
| Weekday | Yes      | Yes           | Yes |
| Holiday | No       | No            | No  |
| Weekday | Yes      | Yes           | Yes |
| Holiday | Yes      | Yes           | Yes |
| Weekend | No       | No            | No  |
| Weekday | Yes      | No            | Yes |
| Holiday | Yes      | Yes           | Yes |
| Weekend | Yes      | Yes           | Yes |
| Weekday | Yes      | Yes           | Yes |
| Holiday | No       | Yes           | Yes |
| Weekend | No       | No            | No  |
| Weekday | Yes      | Yes           | No  |
| Holiday | Yes      | Yes           | Yes |
| Weekday | Yes      | Yes           | Yes |
| Holiday | No       | Yes           | Yes |

Based on the above dataset, let's create the frequency table for each attribute in it.

### Frequency table and Likelihood table for Discount and Buy

Frequency Table

|          |     | Buy |    |
|----------|-----|-----|----|
|          |     | Yes | No |
| Discount | Yes | 19  | 3  |
|          | No  | 3   | 5  |

Likelihood Table

|          |     | Buy   |      |       |
|----------|-----|-------|------|-------|
|          |     | Yes   | No   |       |
| Discount | Yes | 19/22 | 3/8  | 22/30 |
|          | No  | 3/22  | 5/8  | 8/30  |
|          |     | 22/30 | 8/30 | 30    |

### Frequency table and Likelihood table for Free Delivery and Buy

Frequency Table

|               |     | Buy |    |
|---------------|-----|-----|----|
|               |     | Yes | No |
| Free Delivery | Yes | 20  | 3  |
|               | No  | 2   | 5  |

Likelihood Table

|               |     | Buy   |      |       |
|---------------|-----|-------|------|-------|
|               |     | Yes   | No   |       |
| Free Delivery | Yes | 20/22 | 3/8  | 23/30 |
|               | No  | 2/22  | 5/8  | 7/30  |
|               |     | 22/30 | 8/30 | 30    |

### Frequency table and Likelihood table for Day and Buy

Frequency Table

|     |         | Buy |    |
|-----|---------|-----|----|
|     |         | Yes | No |
| Day | Weekday | 6   | 4  |
|     | Weekend | 6   | 4  |
|     | Holiday | 10  | 2  |

Likelihood Table

|     |         | Buy   |      |       |
|-----|---------|-------|------|-------|
|     |         | Yes   | No   |       |
| Day | Weekday | 6/22  | 4/8  | 10/30 |
|     | Weekend | 6/22  | 2/8  | 8/30  |
|     | Holiday | 10/22 | 2/8  | 12/30 |
|     |         | 22/30 | 8/30 | 30    |

From the likelihood table, find probability that a person will buy on holiday (Buy with Yes and Day is Holiday) which is  $10/22 = 0.45$ . Let's find the probability of free delivery and buy, which is  $20/22 = 0.90$ .

Calculate the probability of buy given day is holiday, discount is yes and free delivery is yes. Here let A be the event Buy and B be the event day is holiday, discount is yes and free delivery is yes. Putting all in Bayes' theorem.

**P(A|B)**

$$\begin{aligned}
&= P(\text{Buy} | \text{Day} = \text{Holiday}, \text{Discount} = \text{Yes}, \text{Free Delivery} = \text{Yes}) \\
&= P(\text{Day} = \text{Holiday} | \text{Buy}) * P(\text{Discount} = \text{Yes} | \text{Buy}) * P(\text{Free Delivery} = \text{Yes} | \text{Buy}) \\
&\quad * P(\text{Buy}) / P(\text{Day} = \text{Holiday}) * P(\text{Discount} = \text{Yes}) * P(\text{Free Delivery} = \text{Yes}) \\
&= ((10/22) * (19/22) * (20/22) * (22/30)) / ((12/30) * (22/30) * (23/30)) \\
&= \mathbf{1.1637}
\end{aligned}$$

Calculate the probability of No Buy given day is holiday, discount is yes and free delivery is yes. Here let A be the event No Buy and B be the event day is holiday, discount is yes and free delivery is yes.

**P(A|B)**

$$\begin{aligned}
&= P(\text{No Buy} | \text{Day} = \text{Holiday}, \text{Discount} = \text{Yes}, \text{Free Delivery} = \text{Yes}) \\
&= P(\text{Day} = \text{Holiday} | \text{No Buy}) * P(\text{Discount} = \text{Yes} | \text{No Buy}) * P(\text{Free Delivery} = \text{Yes} | \text{No Buy}) \\
&\quad * P(\text{No Buy}) / P(\text{Day} = \text{Holiday}) * P(\text{Discount} = \text{Yes}) * P(\text{Free Delivery} = \text{Yes}) \\
&= ((2/8) * (3/8) * (3/8) * (8/30)) / ((12/30) * (22/30) * (23/30)) \\
&= \mathbf{0.0417}
\end{aligned}$$

Let's predict likelihood of buying or not buying given day is holiday, discount is yes and free delivery is yes, which is simple sum of above calculated probabilities Buy and No Buy.

$$\text{Sum of probabilities} = 1.1637 + 0.0417 = \mathbf{1.2054}$$

$$\text{So Likelihood of Buy is } P(\text{Buy} | \text{Day} = \text{Holiday}, \text{Discount} = \text{Yes}, \text{Free Delivery} = \text{Yes}) = 1.1637/1.2054 = \mathbf{0.9654}$$

$$\text{So Likelihood of No Buy is } P(\text{No Buy} | \text{Day} = \text{Holiday}, \text{Discount} = \text{Yes}, \text{Free Delivery} = \text{Yes}) = 0.0417/1.2054 = \mathbf{0.0346}$$

We can conclude that people are more likely to buy on holidays if they get discount and free delivery.

## Spam or not Spam e-mail's Classification using Naïve Bayes Technique

Let's explain it using an example to make things clear:

Assume we have a bunch of emails that we want to classify as *spam* or *not spam*.

Our dataset has **15 Not Spam** emails and **10 Spam** emails. Some analysis had been done, and the frequency of each word had been recorded as shown below:

|             | Not Spam | Spam |
|-------------|----------|------|
| Dear        | 8        | 3    |
| Visit       | 2        | 6    |
| Invitation  | 5        | 2    |
| Link        | 2        | 7    |
| Friend      | 6        | 1    |
| Hello       | 5        | 4    |
| Discount    | 0        | 8    |
| Money       | 1        | 7    |
| Click       | 2        | 9    |
| Dinner      | 3        | 0    |
| Total Words | 34       | 47   |

**Note:** Stop Words like "the", "a", "on", "is", "all" had been removed as they do not carry important meaning and are usually removed from texts. The same thing applies to numbers and punctuations.

### exploring some probabilities:

- $P(\text{Dear}|\text{Not Spam}) = 8/34$
- $P(\text{Visit}|\text{Not Spam}) = 2/34$
- $P(\text{Dear}|\text{Spam}) = 3/47$
- $P(\text{Visit}|\text{Spam}) = 6/47$

and so on.

now assume we have the message "Hello friend" and we want to know whether it is a spam or not.

so, using Bayes' Theorem

$$P(\text{Not Spam}|\text{Hello Friend}) = \frac{P(\text{Hello Friend}|\text{Not Spam}) * P(\text{Not Spam})}{P(\text{Hello Friend})}$$

ignoring the denominator

$$P(\text{Not Spam}|\text{Hello Friend}) = P(\text{Hello Friend}|\text{Not Spam}) * P(\text{Not Spam})$$

But, **P(Hello friend | Not Spam) = 0**, as this case (Hello friend) doesn't exist in our dataset, i.e. we deal with single words, not the whole sentence, and the same for **P(Hello friend | Spam)** will be **zero** as well, which in turn will make both probabilities of being a spam and not spam both are **zero**, which has no meaning!!

But wait!! we said that the Naive Bayes assumes that **`the features we use to predict the target are independent`**.

so,

$$P(\text{Hello Friend}|\text{Not Spam}) = P(\text{Hello}|\text{Not Spam}) * P(\text{Friend}|\text{Not Spam})$$

$$P(\text{Not Spam}|\text{Hello Friend}) = P(\text{Hello}|\text{Not Spam}) * P(\text{Friend}|\text{Not Spam}) * P(\text{Not Spam})$$

$$P(\text{Not Spam}|\text{Hello Friend}) = \frac{5}{34} * \frac{6}{34} * \frac{15}{25} = 0.0155$$

now let's calculate the probability of being spam using the same procedure:

$$P(\text{Spam}|\text{Hello Friend}) = \frac{4}{47} * \frac{1}{47} * \frac{10}{25} = 0.00072$$

so, the message "Hello friend" is not a spam.

now, let's try another example:

assume the message "dear visit dinner money money money". It's obvious that it's a spam, but let's see what Naive Bayes will say.

$$P(\text{Not Spam} | \text{dear visit dinner money money money}) \\ = P(\text{dear visit dinner money money money} | \text{Not Spam}) * P(\text{Not Spam})$$

$$P(\text{dear visit dinner money money money} | \text{Not Spam}) = \frac{8}{34} * \frac{2}{34} * \frac{3}{34} * \left(\frac{1}{34}\right)^3 \\ = 3.107 * 10^{-8}$$

$$P(\text{Not Spam} | \text{dear visit dinner money money money}) = 3.107 * 10^{-8} * \frac{15}{25} \\ = 1.864 * 10^{-8}$$

$$P(\text{Spam} | \text{dear visit dinner money money money}) \\ = P(\text{dear visit dinner money money money} | \text{Spam}) * P(\text{Spam})$$

$$P(\text{dear visit dinner money money money} | \text{Spam}) = \frac{3}{47} * \frac{6}{47} * 0 * \left(\frac{7}{47}\right)^3 = 0$$

$$P(\text{Spam} | \text{dear visit dinner money money money}) = 0 * \frac{10}{25} = 0$$

oops!! Naive Bays says that this message is **not a spam**!!!!

This happened because the word "dinner" does not appear in the *spam* dataset, so that **P(dinner | spam) = 0**, hence all other probabilities will have no effect.

This is called the **Zero-Frequency Problem**.

and to solve that we can use **Laplace smoothing**.

**Laplace Smoothing** is a technique for smoothing categorical data. A small-sample correction, or pseudo-count, will be incorporated in every probability estimate. Hence, no probability will be zero. this is a way of regularizing Naive Bayes.

Given an observation  $x = (x_1, \dots, x_d)$  from a multinomial distribution with  $N$  trials and parameter vector  $\theta = (\theta_1, \dots, \theta_d)$ , a "smoothed" version of the data gives the estimator:

$$\hat{\theta} = \frac{x_i + \alpha}{N + \alpha d} \quad (i = 1, \dots, d)$$

where the pseudo-count  $\alpha > 0$  is the smoothing parameter ( $\alpha = 0$  corresponds to no smoothing).

Back to our problem, we are going to pick  $\alpha = 1$ , and 'd' is the number of the unique words in the dataset which is 10 in our case.

$$\hat{\theta} = \frac{x_i + \alpha}{N + \alpha d} \quad \alpha = 1, \text{ and } d = 10$$

$$P(\text{dear visit dinner money money money} | \text{Not Spam}) = \frac{8+1}{34+10} * \frac{2+1}{34+10} * \frac{3+1}{34+10} * \left( \frac{1+1}{34+10} \right)^3$$

$$= 1.19 * 10^{-7}$$

$$P(\text{Not Spam} | \text{dear visit dinner money money money}) = 1.19 * 10^{-7} * \frac{15}{25}$$

$$= 7.144 * 10^{-8} = 0.7144 * 10^{-7}$$

$$P(\text{dear visit dinner money money money} | \text{Spam}) = \frac{3+1}{47+10} * \frac{6+1}{47+10} * \frac{0+1}{47+10} * \left( \frac{7+1}{47+10} \right)^3$$

$$= 0.000000418 = 4.18 * 10^{-7}$$

$$P(\text{Spam} | \text{dear visit dinner money money money}) = 4.18 * 10^{-7} * \frac{10}{25}$$

$$= 1.672 * 10^{-7}$$

Now it's correctly classified the message as spam.

## Application of Naive Bayes

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So how Naive Bayes can be used in real life or industry?. Check out some of the real life scenarios give below:

- **News Categorization**
- **Text classification/ Spam Filtering/ Sentiment Analysis**
- **Fraud Detection**
- **Medical Diagnosis**
- **Weather Prediction**
- **Object identification or Face recognition**
- **Real-time Prediction**
- **Multi-class Prediction**
- **Recommendation Systems.**

## Advantages and Disadvantages of Naive Bayes

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### Advantages

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- It is easy to implement
- It is not very much sensitive to irrelevant features
- It needs less training data
- It can handle both continuous and discrete data
- It is highly scalable with number of predictors and data points
- It is a fast executing algorithm so it can be used in real time predictions
- Requires a small amount of training data. So the training takes less time.
- Handles continuous and discrete data, and it is not sensitive to irrelevant features.
- Very simple, fast, and easy to implement.
- Can be used for both binary and multi-class classification problems.
- Highly scalable as it scales linearly with the number of predictor features and data points.
- When the Naive Bayes conditional independence assumption holds true, it will converge quicker than discriminative models like logistic regression.

### Disadvantages

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- Treats each feature independently, so it will mislead if not used wisely on dependent feature
  - In text classification or filtering, it ignores the order of the words. So this algorithm cannot work well where there is importance of order
  - The assumption of independent predictors/features. Naive Bayes implicitly assumes that all the attributes are mutually independent which is almost impossible to find in real-world data.
  - If a categorical variable has a value that appears in the test dataset, and not observed in the training dataset, then the model will assign it a zero probability and will not be able to make a prediction. This is what we called the **"Zero Frequency problem"**, and can be solved using smoothing techniques.
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